

JAWAD UL HADI

Backend Lead / Architect • AI-First Systems Design

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PROFESSIONAL SUMMARY

Backend Lead Engineer, 7+ years, most of it on multi-tenant SaaS and AI-powered enterprise systems. I work in Node.js, NestJS, and TypeScript over PostgreSQL, MySQL, MongoDB, and Redis, with Python and FastAPI where those fit better, and I own architecture from database schema through deployment. Recent work has been in production AI: Anthropic models, MCP servers, provider abstraction, RAG pipelines, and the resilience patterns that keep them from failing loudly.

CORE SKILLS

- **Architecture & Systems Design:** Microservices architecture, GraphQL API development, event-driven design, multi-tenant SaaS platforms, technical leadership, scalability-reliability trade-off management
- **Backend Engineering:** Node.js, NestJS, TypeScript, Python, FastAPI, REST/GraphQL, PostgreSQL, MySQL, MongoDB, Redis, query and performance optimization
- **AI-First Engineering:** Claude Code, GitHub Copilot, Gemini 2.5 Flash, AI agent workflows for scaffolding/implementation/testing, RAG pipelines, MCP servers, LLM provider integration (OpenAI, Gemini, Anthropic)
- **Reliability & Delivery:** CI/CD (GitHub Actions), Jest, automated testing strategy, OAuth 2.0/JWT security, AWS/GCP, Docker/Kubernetes

PROFESSIONAL EXPERIENCE

Backend Lead / Senior Software Engineer

MicroAgility Services (Pvt) Ltd. | Islamabad, Pakistan

Jan 2024 – Present

- Cut AI integration complexity by 60% and eliminated user-facing failures during provider outages by designing a provider-agnostic microservices architecture — a routing/gateway service fronting pluggable OpenAI, Gemini, and Anthropic adapters — with automatic failover behind a single abstraction.
- Achieved zero user-facing AI failures during live LLM outages, measured across all production incidents, by designing a three-tier resilience pattern (Retry → RAG Fallback → Rule-Based Floor) and defining it as the team's standard failure-handling architecture.
- Reduced dashboard response times from 12s to under 2s (83% improvement) by owning end-to-end PostgreSQL and MongoDB query and performance optimization, indexing strategy, and MeiliSearch rollout.
- Sustained sub-second API response times under thousands of concurrent async jobs by designing an event-driven processing architecture on BullMQ and Redis and setting the technical direction for high-throughput workloads.
- Shortened release cycles and lifted developer productivity, measured by faster time-to-deploy, by building GitHub Actions CI/CD pipelines with automated test coverage as a standing testing strategy.
- Held zero cross-tenant data exposure incidents across the platform's lifetime by architecting secure REST and GraphQL APIs on OAuth 2.0/JWT and enforcing tenant isolation as a non-negotiable design standard.
- Delivered a complete timesheet module end-to-end, measured by full Jest unit-test coverage at handoff, using GitHub Copilot as the primary implementation tool under the same review and correctness bar as hand-written code.
- Built an internal ATS platform, measured by shipped production features, using Claude Code as an AI pair-programmer for scaffolding and Gemini 2.5 Flash for document parsing and the conversational assistant, while retaining full ownership of architecture and review.
- Raised engineering consistency across the team, measured by review-standard adoption, by mentoring backend engineers through architecture and code review covering both AI-assisted and hand-written code.

- Authored “Designing for AI Failure,” a technical case study documenting the platform's three-tier resilience architecture with system diagrams and a reversible vector-store ADR, used internally to align technical direction and onboard engineers.

Backend Software Engineer

MicroAgility Services (Pvt) Ltd. | Islamabad, Pakistan

Mar 2022 – Jan 2024

- Maintained 99.9% uptime through zero-downtime schema migrations by designing a migration process that ran against live multi-tenant production databases.
- Improved data retrieval speed by ~40% by refactoring REST and GraphQL services serving thousands of users across multiple tenants.
- Cut manual onboarding effort for new enterprise tenants by automating the tenant's deployment and migration pipeline.

Software Engineer

Market Icon | Rawalpindi, Pakistan

Jan 2018 – Mar 2022

- Eliminated idle infrastructure cost by designing serverless AWS Lambda/API Gateway systems on an event-driven architecture instead of always-on servers.
- Enabled secure third-party integrations (CRM, payments) by building APIs across Express.js, FastAPI, and Django REST Framework.

EDUCATION & CERTIFICATIONS

B.S. Computer Science — Government College University, Faisalabad, 2018

Certifications: AI/LLM Systems, Cloud Architecture, and Backend Engineering credentials from IBM, Microsoft, Google, and Anthropic. Full credential list available in [Portfolio](#).

ADDITIONAL INFORMATION

- **Languages:** English (professional working proficiency), Urdu (native)
- **Availability:** Full-time; open to relocation anywhere; remote or on-site.